Gravity Model Simulation of the International Trade Network using a Preferentially Selected Network

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Abstract

We introduce a model to simulate Gross Domestic Products (GDP) from the International Trade Network (ITN) data. For this simulation, we apply two complex network concepts; the Preferentially Selected Network (PSN) and Linear Gravity Transportation Model (GTM), where PSN is created by removing links with smaller contributions. On this network, we simulate sales and GDP of each node on the ITN using the Linear GTM.

Keyword: Preferentially Selected Network, Gravity model, International Trade Network

1 International Trade Network (ITN)

Researchers have explored ITN structure from 2003 and found Gross Domestic Product (GDP) is a hidden factor of the network [1, 2].

In this study, we reproduce sales and GDP of each country on the ITN through simulation. We introduce two complex network concepts called Preferentially Selected Network (PSN) and Linear Gravity Transportation Model (GTM).

2 Network concepts "PSN and GTM"

PSN is a simple network produced from the original network in which links with small contributions are removed by the following rule: "For all the nodes, select top n^{th} weight links for both in-flow and out-flow and cut off all other links."

We also use GTM known in the social network research [3]. In the case of ITN sales, we apply a linearized version of GTM. By tuning parameters of the model, we can reproduce the distribution of the amount of exports and GDP for all countries as shown in Figure 1.

3 Conclusion

We estimated exports and GDPs by solving the linear equation of GTM for given network structure, PSN, and the results became very close to the real values.

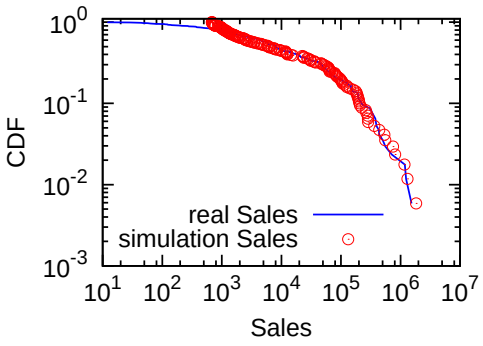

Figure1: Exports (Sales) distribution estimated from simulation (n=7) and real data.

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